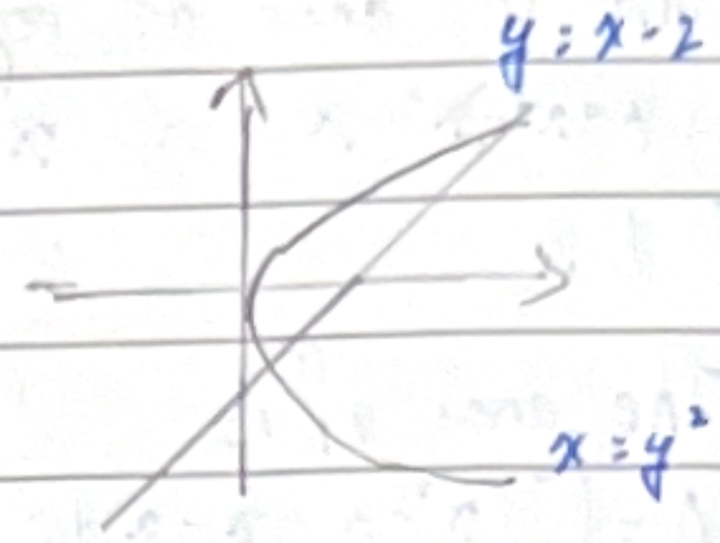
1. Find the area of region bounded by $x=y^2$ and $x-y=2$ $\cancel{x=y+2}$



(1) $\begin{cases} x=y+2 \\ x=y^2 \end{cases}$ $\Rightarrow y^2-y-2=0$ $\Rightarrow y=2$ or -1

The area is

$A=\int_{-1}^{2}((y+2)-y^2)dy$

$=(-\frac{1}{3}y^3+\frac{1}{2}y^2+2y)\Big|_{-1}^{2}=-\frac{8}{3}+2+4-(\frac{1}{3}+\frac{1}{2}-2)=\frac{9}{2}$

(2) $\begin{cases} x=y+2 \\ x=y^2 \end{cases}$ $\Rightarrow y=-1$ or 2

The area is

$A=\int_{-1}^{2}(y+2-y^2)dy$

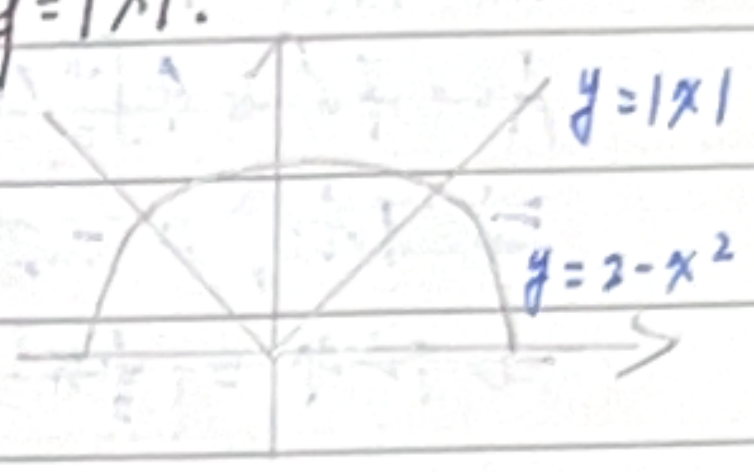
$=(-\frac{1}{3}y^3+\frac{1}{2}y^2+2y)\Big|_{-1}^{2}=-\frac{8}{3}+2+4-(\frac{1}{3}+\frac{1}{2}-2)=\frac{9}{2}$

(3) $\begin{cases} x=y+2 \\ x=y^2 \end{cases}$ $\Rightarrow y=2$ or -1

The area is

$A=\int_{-1}^{2}(y+2-y^2)dy=(-\frac{1}{3}y^3+\frac{1}{2}y^2+2y)=\frac{9}{2}$

2. Find the area of the region bounded by $y=2-x^2$ and $y=|x|$.



(1) $\begin{cases} y=2-x^2 \\ y=|x| \end{cases}$ $\Rightarrow x=-1$ or 1

The area is

$A=2\int_{0}^{1}(2-x^2-x)\,dx=2(2x-\frac{1}{3}x^3-\frac{1}{2}x^2)\Big|_0^1=2(2-\frac{1}{3}-\frac{1}{2})=\frac{7}{3}$

(2) $\begin{cases} y=2-x^2 \\ y=|x| \end{cases}$ $\Rightarrow x=-1$ or $x=1$

The area is

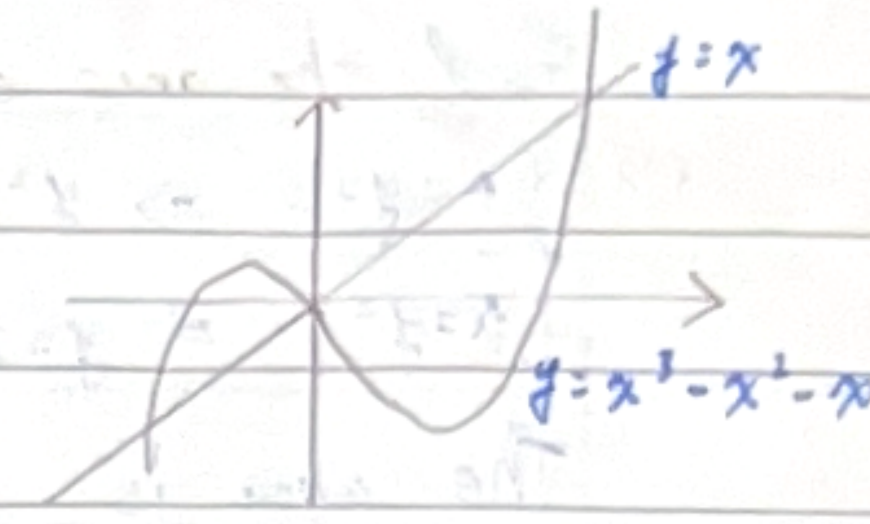
$A=2\int_{0}^{1}(2-x^2-(x))dx=2[2x-\frac{1}{2}x^2-\frac{1}{3}x^3]\Big|_0^1=2(2-\frac{1}{2}-\frac{1}{3})=\frac{7}{3}$

(3) $\begin{cases} y=2-x^2 \\ y=|x| \end{cases}$ $\Rightarrow x=-1$ or 1

The area is

$A=2\int_{0}^{1}(2-x^2-(x))dx=2(2x-\frac{1}{2}x^2-\frac{1}{3}x^3)\Big|_0^1=2(2-\frac{1}{2}-\frac{1}{3})=\frac{7}{3}$

3. Find the area of region bounded by $y=x^3-x^2-x$ and $y=x$.



(1) $\begin{cases} y=x^3-x^2-x \\ y=x \end{cases} \Rightarrow x^3-x^2-2x=0$

$x=-1$ or 0 or 2

The area of is $\quad -\frac{1}{4}x^3+\frac{1}{3}x^3+x^2$

$A=\int_{-1}^{0}(x^3-x^2-x-x)\cancel{dx}+\int_0^2(x-x^3+x^2+x)dx$

$=(-\frac{1}{4}x^4-\frac{1}{3}x^3-x^2)\Big|_{-1}^{0}+(x^2+\frac{1}{3}x^3-\frac{1}{4}x^4)\Big|_0^2$

$=\frac{5}{12}+\frac{8}{3}=\frac{37}{12}$

(2) $\begin{cases} y=x^3-x^2-x \\ y=x \end{cases} \Rightarrow x=-1$ or $x=0$ or $x=2$

The area is

$A=\int_{-1}^{0}(x^3-x^2-x-(x))dx+\int_0^2(x-(x^3-x^2-x))dx$

$=(-\frac{1}{4}x^4-\frac{1}{3}x^3-x^2)\Big|_{-1}^{0}+(x^2+\frac{1}{3}x^3-\frac{1}{4}x^4)\Big|_0^2$

$=\frac{5}{12}+\frac{8}{3}=\frac{37}{12}$

(3) $\begin{cases} y=x^3-x^2-x \\ y=x \end{cases} \Rightarrow x=-1$ or $x=0$ or $x=2$

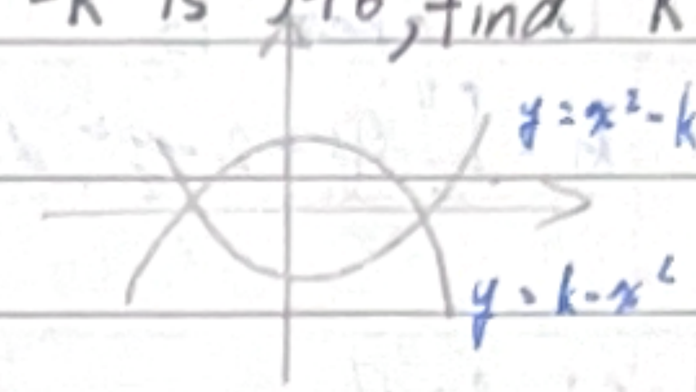
The area is

$A=\int_{-1}^{0}(x^3-x^2-x-(x))dx+\int_0^2(x-(x^3-x^2-x))dx$

$=(\frac{1}{4}x^4-\frac{1}{3}x^3-\frac{2}{2}x^2)\Big|_{-1}^{0}+(x^2+\frac{1}{3}x^3-\frac{1}{4}x^4)\Big|_0^2$

$=\frac{5}{12}+\frac{8}{3}=\frac{37}{12}$

4. If the area of the region bounded by $y=k-x^2$ and $y=x^2-k$ is 576, find k.



(1) $\cancel{\begin{cases} y=k-x^2 \\ y=x^2-k \end{cases} \Rightarrow x^2-k+x^2-k=0 \quad x=\sqrt{k} \text{ or } -\sqrt{k}}$

~~The area is~~

$\cancel{A=576=4\int_0^{\sqrt{k}}(x^2-k-k+x^2)dx=4(-2kx+\frac{2}{3}x^3)\Big|_0^{\sqrt{k}}}$

$\cancel{\Rightarrow 144=-2k\sqrt{k}+\frac{2}{3}k\sqrt{k}=}$

$\cancel{\Rightarrow 144=(\frac{1}{3}x^3-kx)\Big|_0^{\sqrt{k}} \Rightarrow 144=\frac{1}{3}(\sqrt{k})^3-(\sqrt{k})^3 \Rightarrow 216=-(\sqrt{k})^3 \Rightarrow k=36}$

(2) $\begin{cases} y=k-x^2 \\ y=x^2-k \end{cases} \Rightarrow x=\sqrt{k}$ or $-\sqrt{k}$

The area is

$A=576=4\int_0^{\sqrt{k}}(k-x^2)dx \Rightarrow (kx-\frac{1}{3}x^3)\Big|_0^{\sqrt{k}}=144 \Rightarrow k^{\frac{3}{2}}-\frac{1}{3}k^{\frac{3}{2}}=144 \Rightarrow k^{\frac{3}{2}}=216 \Rightarrow k=36$

(3) $\begin{cases} y=k-x^2 \\ y=x^2-k \end{cases} \Rightarrow x=-\sqrt{k}$ or $\sqrt{k}$

The area is

$A=576=4\int_0^{\sqrt{k}}(k-x^2)dx=4(kx-\frac{1}{3}x^3)\Big|_0^{\sqrt{k}}$

$\Rightarrow 144=k^{\frac{3}{2}}-\frac{1}{3}k^{\frac{3}{2}}$

$\Rightarrow k^{\frac{3}{2}}=216 \Rightarrow k=36$

(1) $\begin{cases} y=k-x^2 \\ y=x^2-k \end{cases} \Rightarrow x=-\sqrt{k}$ or $\sqrt{k}$

The area is

$A=576=4\int_0^{\sqrt{k}}(k-x^2)dx=4(kx-\frac{1}{3}x^3)\Big|_0^{\sqrt{k}}$

$\Rightarrow 144=k^{\frac{3}{2}}-\frac{1}{3}k^{\frac{3}{2}}$

$\Rightarrow k^{\frac{3}{2}}=216 \Rightarrow k=36$

7-2 Volumes

切割大量的A 从 $x=a \sim x=b$

$V=\int_a^b A(x)\,dx$

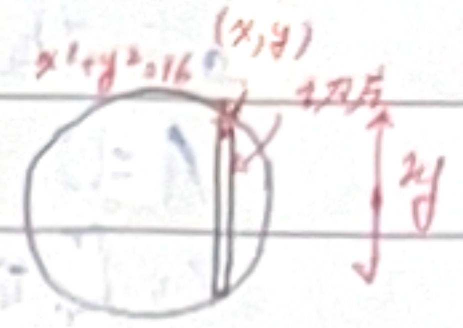
正三角形

先算切片面积，再算体积

Let S be a solid with a circular base of radius 6. If each parallel cross section perpendicular to the base is an equilateral triangle, find the volume of S.

$x^2+y^2=r^2$ $\quad \frac{\sqrt{3}a^2}{4}$

(1) Let $x^2+y^2=36$ be the base of S and $\overline{AB}$ is the base of the cross triangle. Then the area of the cross triangle is



$A(x)=\frac{\sqrt{3}}{4}\overline{AB}^2=\frac{\sqrt{3}}{4}(2y)^2=\sqrt{3}(36-x^2)$

$V=\int_{-6}^{6}A(x)\,dx=2\int_0^6\sqrt{3}(36-x^2)\,dx=2\sqrt{3}\left(36x-\frac{1}{3}x^3\right)\Big|_0^6=2\sqrt{3}(216-72)=288\sqrt{3}$

(2) Let $x^2+y^2=36$ be the base of S and $\overline{AB}$ is the base of the cross triangle.
Then the area of the cross triangle is

$A(x)=\frac{\sqrt{3}}{4}\overline{AB}^2=\frac{\sqrt{3}}{4}(2y)^2=\sqrt{3}(36-x^2)$

$V=\int_{-6}^{6}A(x)\,dx=2\int_0^6\sqrt{3}(36-x^2)\,dx=2\sqrt{3}\int_0^6(36-x^2)\,dx$

$=2\sqrt{3}\left(36x-\frac{1}{3}x^3\right)\Big|_0^6=2\sqrt{3}(216-72)=288\sqrt{3}$

(3) Let $x^2+y^2=36$ be the base of S and $\overline{AB}$ is the base of the cross triangle.
Then the area of the cross triangle is

$A(x)=\frac{\sqrt{3}}{4}\overline{AB}^2=\frac{\sqrt{3}}{4}(2y)^2=\sqrt{3}(36-x^2)$

The volume of S is

$V=2\int_0^6 A(x)\,dx=2\int_0^6\sqrt{3}(36-x^2)\,dx=2\sqrt{3}\left(36x-\frac{1}{3}x^3\right)\Big|_0^6$

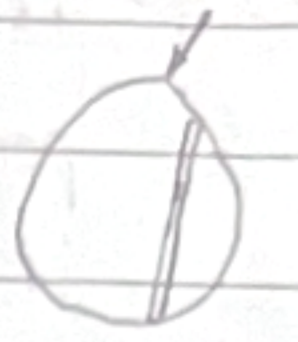
$=2\sqrt{3}(216-72)=288\sqrt{3}$

2nd week No.: Date:

Disk Method 悬转体 (like 圆的πr^2) 对悬转轴axis切

$$V=\int_c^d \pi (g(y))^2\,dy$$

Prove the volume of a sphere of radius r is $\frac{4}{3}\pi r^3$



(1) Let $y=\sqrt{r^2-x^2}$ be the function of upper semi-circle of the radius r

$A(y)=\pi y^2=\pi(\sqrt{r^2-x^2})^2$

$V=\int_{-r}^{r}\pi(\sqrt{r^2-x^2})^2\,dx=2\pi\int_0^r(r^2-x^2)\,dx=2\pi\left(r^2x-\frac{1}{3}x^3\right)\Big|_0^r=2\pi\left(r^3-\frac{1}{3}r^3\right)=\frac{4}{3}\pi r^3$

(2) Let $y=\sqrt{r^2-x^2}$ be the function of upper semi-circle of the radius r

The area is

$A(y)=\pi y^2=\pi(\sqrt{r^2-x^2})^2$

The volume is

$V=\int_{-r}^{r}\pi(\sqrt{r^2-x^2})^2\,dx=2\pi\int_0^r(r^2-x^2)\,dx=2\pi\left(r^2x-\frac{1}{3}x^3\right)\Big|_0^r$

$=2\pi\left(r^3-\frac{1}{3}r^3\right)=\frac{4}{3}\pi r^3$

(3) Let $y=\sqrt{r^2-x^2}$ be the function of upper semi-circle of the radius r

The Volume is

$V=\int_{-r}^{r}\pi y^2\,dx=2\pi\int_0^r(r^2-x^2)\,dx=2\pi\left(r^2x-\frac{1}{3}x^3\right)\Big|_0^r$

$=2\pi\left(r^3-\frac{1}{3}r^3\right)=\frac{4}{3}\pi r^3$